Journal-Based Mental Health Risk Assessment via Integrated Emotion Classification and Risk Flag Detection Using BERT

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1 Introduction

Mental health disorders represent a growing global health crisis, with the World Health Organization (WHO) estimating that in 2019 approximately one in eight people worldwide were living with a mental health condition. The COVID-19 pandemic further intensified this issue [1]. A Chinese study [2] carried out in Wuhan, China showed that 96.2% of clinically stable COVID-19 inpatients reported significant post-traumatic stress symptoms prior to discharge, which would suggest potential impacts on quality of life. Since the beginning of the outbreak, based on the experience of past pneumonia outbreaks that brought themes of fear, uncertainty, and stigmatisation, it was hypothesized that the implementation of mental health assessment, support, treatment, and services were crucial goals for the health response [3]. People with mental disorders report that the disorders carry a lot of difficulty even in small daily activities and meeting demands at work [4]. Many factors affect people's mental health, but when left untreated, severe mental disorders may lead to suicidal thoughts or behaviors [5], and each year, one death by suicide occurs every 40 seconds (World Health Organization, 2021). The American Foundation for Suicide Prevention (AFSP) has found that suicide factors fall under three categories: health factors, environmental factors, and historical factors [6]. Good support programs can lessen most of these factors and prevent many suicidal tragedies.

Detection of mental health problems, especially suicidal tendencies, has become the focus of many researchers due to an increase in suicide rates and mental health problems in general. A substantial amount of research includes looking for ways to help mental health professionals assess risks by using Artificial Intelligence (AI). Use of AI in mental health is often frowned upon—mostly by mental health professionals. The reasons behind this hesitancy are doubts about preserving the therapeutic alliance, fears of dehumanization, worries about data privacy and algorithmic bias, and the practical challenges that come with integrating AI tools into clinical workflows [7]. However, as AI integration became more common even in daily tasks [8], its methods for use in therapy have also been expanded. These methods range from real-time assistance to mental health professionals during therapy sessions [?] to chatbots that help the patient self-disclose sensitive topics to their counselors and act as passive mediators in therapy [10]. A systematic review of machine learning (ML) models in suicide prevention by Bernert et al. explained how ML-based models consistently outperformed clinician-based predictions in risk assessment [11].

Aside from their impact in counselor-led therapy, ML approaches in suicide prediction have been more prevalent in the online world. For instance, Google has developed the "Google Depression Screening Tool"[1] which screens for depression-related terms in users' search queries and prompts them with the Patient Health Questionnaire-9 (PHQ-9), then provides guidance on seeking professional help if necessary based on the responses. Online social networks (OSNs) have become a modern place for venting to strangers and because of this new medium of expression, a need for new ways of detecting mental health problems has emerged [12]. They have shown that despite the fact that OSNs exhibit high potential as data sources for detecting mental health problems, they cannot substitute traditional detection methods based on face-to-face interviews, self-reporting, or questionnaire distribution [13].

Conversational Artificial Intelligence (CAI) has gained popularity over the recent years and while it can provide some harmful suggestions due to a lack of ability to grasp the full context [14], it can be useful in helping both patients with self-expression [10] and counselors by providing real-time suggestions [?]. The growing use of chatbots as conversational partners raises significant ethical concerns, particularly given users' tendency to overshare in detail [15]. This concern over AI companions was tragically illustrated by recent news reports covering deaths by suicide of young people influenced by harmful chatbot suggestions and reports of individuals convinced by AI-driven conspiracies that led to murder.

Given the notable benefits and serious risks that come with AI integration into mental healthcare, there is a vital need for responsible systems designed with help and safety in mind. This includes protecting users' privacy while ensuring they receive urgent help in critical situations. This study aims to meet these needs by developing an AI model for detecting emotional distress and mental health risks through a digital journal. The ultimate goal is integration into a mobile application that sends timely notifications to a trusted individual—preferably a mental health professional—when high risk of self-harm or suicidal tendencies is detected. The focus of this study is therefore the predictive model, while the mobile application represents a future implementation effort.

2 Literature Review

Related work is considered in the following categories: social media content for detecting mental health problems, impact of CAI on mental health, digital

[1] `https://health.google/mental-health/`

journals and sentiment analysis, and app-based mental health support.

2.1 OSN Content for Detecting Mental Health Problems

Numerous datasets derived from OSNs are used in NLP tasks for emotion and suicide risk detection. Li et al. assembled a dataset expanding coverage of suicidal posts with fine-grained risk annotations [16]. These annotations, approved by psychology experts, include indicator, ideation, behaviour, and attempt [17].

Vongtangton and Goebert designed a multimodal predictive model using Instagram data from 142 users and found distinct linguistic and visual markers associated with suicide risk [18]. Their findings motivate the inclusion of multimodal signals in future work.

2.2 Impact of CAI on Mental Health

ELIZA, developed by Joseph Weizenbaum in the 1960s, laid the foundation for modern chatbots [19]. Waszak conducted an early post-deployment analysis of ChatGPT and highlighted safety limitations related to context retention [14]. Schoene and Canca compared six LLMs on self-harm safety compliance, finding significant disparities between free and paid models [21].

Therapeutic chatbots such as Woebot and Wysa employ CBT-based conversational strategies and have demonstrated reductions in depression symptoms [22, 23]. Mishra et al. found AI companions reduced loneliness among urban users [24]. Lee et al. showed that chatbot-mediated disclosure improved counselor interactions [10].

2.3 Digital Journals and Sentiment Analysis

Digital journals and emotion-tracking applications have become popular tools for monitoring emotional well-being [25]. The EMO Diary project integrates NLP-based emotion analysis into daily journaling [26]. Transformer-based models such as BERT have significantly advanced emotion recognition [27]. Ramakrishnan and Babu demonstrated improved performance on imbalanced emotion datasets using BERT with clipped asymmetric loss [28]. Ethical considerations surrounding Emotional AI are discussed by Chavan et al. [29].

2.4 App-Based Mental Health Support

Numerous applications support emotional tracking and suicide prevention, including iBobbly [30], How We Feel [31], Headspace[2], and Calm[3]. These tools often integrate AI-driven personalization. Suicide prevention features include coping exercises, safety planning, and emergency contact support [32]. EMO Diary exemplifies integration of daily routines with mental health monitoring [26].

[2] `https://www.headspace.com/headspace-meditation-app/`
[3] `https://www.calm.com/`

3 Datasets

3.1 Six-Label Emotion Dataset

Training on six emotion labels was performed using a large-scale dataset consisting of more than 450,000 text instances. The final dataset is constructed by combining multiple sources: one primary dataset from Kaggle, four auxiliary Kaggle datasets, and one custom-generated dataset.

3.1.1 Main Dataset (All Six Labels)

The primary dataset was obtained from Kaggle[4] and contains the following properties:

- **Labels**: joy, sadness, anger, fear, surprise, love
- **Size**: 422,566 samples
- **Columns**: `text`, `label`

Preparation Instructions

1. The main dataset file is renamed to `dataset_6labels.csv`, and placed into `data/raw/`.
2. Then four auxiliary auxiliary datasets (listed below) are downloaded, placed in `data/raw/`, and named as instructed below.
3. By running the notebook `../notebooks/merge_love_surprise.ipynb` the file `love_surprise_bonus.csv` is produced into `data/raw/`.
4. Running the preprocessing script:

   ```
   python -m scripts.clean_dataset
   ```

   produces two cleaned datasets:

   - `data/processed/dataset_6labels_clean.csv` (approximately 422k samples)
   - `data/processed/dataset_6labels_clean_more.csv` (more than 450k samples)

3.1.2 Auxiliary Kaggle Datasets (Love and Surprise Only)

To address class imbalance in the *love* and *surprise* categories, four additional datasets were incorporated.

[4] `https://www.kaggle.com/datasets/kushagra3204/sentiment-and-emotion-analysis-dataset`

Extra Dataset 1 Source: Kaggle[5]
File path: `data/raw/extra_dataset_Praveen.csv`

- love: 1,641 instances
- surprise: 719 instances

Extra Dataset 2 Source: Kaggle[6]
File path: `data/raw/extra_dataset_PashupatiGupta.csv`

- love: 3,842 instances
- surprise: 2,946 instances

Extra Dataset 3 Source: Kaggle[7]
File path: `data/raw/extra_dataset_SimaAnjali.csv`

- love: 39,553 instances
- surprise: 6,954 instances

Extra Dataset 4 Source: Kaggle[8]
File path: `data/raw/extra_dataset_BryanHuerta.csv`

- love: 26 instances
- surprise: 31 instances

3.1.3 Custom Surprise Dataset

In addition to publicly available data, a custom dataset was created to further enrich the *surprise* class. A total of 663 instances were generated using large language models (ChatGPT and DeepSeek), guided by fine-grained sublabels such as confusion and excitement and varying text lengths.

All custom instances are labeled exclusively as *surprise* and are stored in the file `more_surprise.csv` within `data/raw/`.

4 Training and Inference Pipeline

The proposed system follows a strict separation between learning, inference, and interpretive explanation. All statistical learning is confined to the training phase, while inference-time mechanisms are designed to manage semantic ambiguity and uncertainty without altering learned parameters.

[5] `https://www.kaggle.com/datasets/praveengovi/emotions-dataset-for-nlp`
[6] `https://www.kaggle.com/datasets/pashupatigupta/emotion-detection-from-text`
[7] `https://www.kaggle.com/datasets/simaanjali/emotion-analysis-based-on-text/data`
[8] `https://www.kaggle.com/datasets/bryanhuerta/social-media-sentiment-data`

4.1 Training Phase

During training, the model learns shared representations and task-specific decision boundaries using supervised objectives only:

1. Input text is tokenized and encoded using a shared pretrained BERT encoder.
2. The encoder output is passed to task-specific heads for emotion recognition and risk detection.
3. Emotion and risk logits are computed independently.
4. Standard loss functions are optimized jointly.
5. No lenient decoding rules, lexicons, or interpretation constraints are applied during training.

4.2 Inference Phase

At inference time, the system augments raw model outputs with structured interpretation mechanisms:

1. Emotion logits are converted to probabilities via softmax and ranked.
2. Lenient decoding rules are applied to semantically overlapping emotions (*love*, *surprise*).
3. Risk probabilities are produced by learned classifiers and evaluated against empirically optimized thresholds.
4. Lexical evidence is used only to support interpretive disclosure, not to influence prediction.
5. Themes and subthemes are inferred via emotion-weighted lexical compatibility.

5 Training Configuration and Experimental Setup

The model was trained in 2 epochs with multitask risk supervision enabled, accumulation step-size 2, learning rate 2.8×10^{-5} early stopping activated (patience = 1), dropout rate set to 0.2, checkpoint saving enabled, and model version `v1_1`. Gradient accumulation was used to achieve an effective batch size of 32 while maintaining memory efficiency.

To address class imbalance in emotion recognition, class-weighted loss was applied with the following weights:

$$\{\text{anger} : 1.27,\ \text{fear} : 1.52,\ \text{joy} : 0.53,\ \text{love} : 1.32,\ \text{sadness} : 0.62,\ \text{surprise} : 3.34\}.$$

These weights reflect inverse frequency scaling and emphasize underrepresented emotions, particularly *surprise*.

6 Training Dynamics and Convergence Behavior

Across two epochs, the model exhibited stable and healthy convergence. Training loss decreased substantially from 0.1670 to 0.0708, while validation loss decreased more modestly from 0.1466 to 0.1377. This pattern indicates continued learning with diminishing returns rather than overfitting. The absolute gap between training and validation loss in epoch 2 reflects expected behavior in high-capacity pretrained transformers nearing representational saturation, not pathological divergence. No instability, collapse, or validation degradation was observed.

Early stopping was not triggered, and checkpoints were saved at both epochs, with the final model stored after epoch 2.

7 Risk Detection Performance

Risk heads converged rapidly, with metrics remaining highly stable across epochs. Changes in F1-score were ≤ 0.002, AUROC values were effectively unchanged, and precision–recall trade-offs were minor and plausible. This indicates that risk detection benefits strongly from pretrained representations and converges early in training.

Final risk performance metrics are summarized as follows:

- **Depression**: F1 = 0.954, AUROC = 0.990, Precision = 0.944, Recall = 0.964, Support = 4768
- **Self-harm**: F1 = 0.993, AUROC = 0.9999, Precision = 1.000, Recall = 0.986, Support = 592
- **Suicidal**: F1 = 0.985, AUROC = 0.997, Precision = 0.999, Recall = 0.972, Support = 3194
- **Grief**: F1 = 0.983, AUROC = 0.998, Precision = 0.997, Recall = 0.968, Support = 820

High AUROC values are expected and non-suspicious in this setting. AUROC measures ranking quality rather than thresholded decisions; the fact that precision and recall are not perfect confirms the absence of leakage or trivial memorization.

Class balance diagnostics further support this interpretation:

$$\text{Depression: } 5846^{-}/4768^{+}, \quad \text{Self-harm: } 600^{-}/592^{+}, \quad \text{Suicidal: } 3600^{-}/3194^{+}, \quad \text{Grief: } 900^{-}/820^{+}.$$

These datasets are sufficiently large and balanced to support stable and meaningful evaluation.

8 Emotion Classification Methodology

Emotion predictions are produced as a probability distribution over the six classes

$$\mathbf{p} = P(\text{emotion} \mid \text{text}) = [p_{\text{joy}}, p_{\text{sadness}}, p_{\text{anger}}, p_{\text{fear}}, p_{\text{surprise}}, p_{\text{love}}],$$

where

$$\sum_i p_i = 1,$$

and the probabilities are obtained by applying the *softmax* function to the model's output logits.

The predicted emotion for a given input is the class with the highest probability:

$$\hat{y} = \arg\max_i p_i, \quad i \in \{\text{joy, sadness, anger, fear, surprise, love}\}.$$

Evaluation is performed using standard multi-class metrics such as accuracy, macro-F1, and weighted-F1:

$$\text{Accuracy} = \frac{\text{Number of correct predictions}}{\text{Total number of samples}},$$

$$\text{Macro-F1} = \frac{1}{L} \sum_{i=1}^{L} \text{F1}_i,$$

where $L = 6$ is the number of emotion classes, and F1_i is the F1-score for class i.

9 Emotion Recognition Performance

Emotion classification performance is realistic and consistent with GoEmotions-style benchmarks. Overall accuracy reached 0.913, with macro F1 of 0.891 and weighted F1 of 0.915. No emotion class approaches perfect performance. Predictably, *love* and *surprise* remain the weakest classes, and *fear* exhibits reduced recall. This non-uniformity further confirms the absence of metric inflation or label leakage.

10 Threshold Optimization and Interpretation

Risk predictions are produced as probabilities

$$p = P(\text{risk} = 1 \mid \text{text}),$$

which are converted to binary decisions using a threshold τ:

$$\hat{y} = \begin{cases} 1 & \text{if } p \geq \tau, \\ 0 & \text{otherwise.} \end{cases}$$

Per-risk threshold sweeping was performed on validation data to maximize F1-score:

$$\tau^* = \arg\max_{\tau \in [0,1]} \mathrm{F1}(\hat{y}(\tau), y).$$

The resulting optimal thresholds were:

$$\{\text{depression} : 0.9,\ \text{self-harm} : 0.09,\ \text{suicidal} : 0.08,\ \text{grief} : 0.06\}.$$

These thresholds are not arbitrary. They reflect differences in lexical clarity, semantic ambiguity, class prevalence, and error cost. Depression receives a highly conservative threshold due to its diffuse linguistic nature and high false-positive cost, whereas self-harm and suicidal ideation receive permissive thresholds to minimize false negatives in safety-critical contexts. Grief exhibits a low threshold because its language is lexically explicit and temporally grounded, making even low-probability predictions reliable.

Importantly, high AUROC combined with low thresholds is expected: AUROC evaluates ranking quality, while threshold sweeping corrects for calibration compression without retraining.

11 Interpretive and Ethical Framing

Thresholds define statistical operating points, not clinical urgency. Depression is treated conservatively to avoid diagnostic framing, while mid-confidence outputs ($0.5 \leq p < 0.9$) are interpreted as ambiguous linguistic signals rather than categorical claims. This explicit acknowledgment of uncertainty improves transparency and ethical alignment.

Risk categories are ontologically asymmetric:

Risk	**Temporal Nature**	**Ambiguity**	**Threshold**
Depression	Chronic	High	High
Grief	Episodic	Medium	Low
Self-harm	Behavioral	Low	Low
Suicidal	Intent-based	Very Low	Low

Low thresholds for grief and suicidal risk arise for opposite reasons: detectability versus harm minimization. This distinction is critical and explicitly documented.

12 Implications for Inference and Temporal Modeling

Inference is conceptualized as a two-layer process:

- **Prediction layer**: numerical beliefs produced by the model.
- **Interpretation layer**: context-aware communication of those beliefs.

The system is not intended to diagnose or replace professional judgment. Instead, it summarizes linguistic risk signals over time. Each journal entry already yields a timestamped multivariate signal (emotion distribution and risk probabilities), forming an implicit time series.

A simple, interpretable temporal aggregation strategy is therefore appropriate:

1. Daily aggregation of signals (e.g., mean or max).
2. Temporal smoothing via rolling or exponentially decayed averages.
3. Interpretation via descriptive bands (e.g., intermittent vs. persistent depressive language).

This approach is transparent, privacy-preserving, and clinically aligned. It avoids premature adoption of opaque learned temporal models and supports ethical deployment.

Conclusion from training phase. The model converges rapidly, exhibits stable multitask behavior, and achieves strong yet realistic performance. Risk detection saturates early, emotion supervision stabilizes shared representations, and per-task threshold optimization yields principled operating points. High AUROC values reflect effective ranking rather than metric inflation. The overall design aligns with best practices in calibrated, safety-aware mental health NLP and provides a solid foundation for interpretable temporal analysis.

13 Joint Emotion Recognition and Risk Detection

13.1 Task Formulation

The model jointly addresses two related but structurally distinct NLP tasks:

1. **Emotion recognition**, framed as single-label classification over

$$\mathcal{E} = \{\text{joy}, \text{love}, \text{sadness}, \text{fear}, \text{surprise}, \text{anger}\}$$

2. **Risk detection**, framed as multi-label binary classification over

$$\mathcal{R} = \{\text{depression}, \text{selfharm}, \text{suicidal}, \text{grief}\}$$

While both tasks operate on the same input text, they maintain independent decision boundaries and error characteristics.

13.2 Model Architecture

13.2.1 Shared Encoder

Let x denote an input text sequence. A pretrained BERT encoder parameterized by θ produces a contextual embedding:

$$h = f_\theta(x), \quad h \in R^d$$

This representation is shared across all downstream tasks.

13.2.2 Emotion Classification Head

Emotion prediction is performed using a softmax classifier:

$$\mathbf{p} = \mathrm{softmax}(W_e h + b_e)$$

where $\mathbf{p} \in R^{|\mathcal{E}|}$, $p_k = P(e_k \mid x)$, and $\sum_k p_k = 1$.

13.2.3 Risk Classification Heads

Each risk label is modeled independently using a sigmoid-activated binary classifier:

$$P(r_j \mid x) = \sigma(W_j h + b_j), \quad r_j \in \mathcal{R}$$

This formulation permits multiple concurrent risk activations.

13.3 Training Objective

13.3.1 Emotion Loss

Emotion recognition is optimized using categorical cross-entropy:

$$\mathcal{L}_{\text{emotion}} = -\log p_y$$

where $y \in \mathcal{E}$ is the gold emotion label.

13.3.2 Risk Loss

Risk detection is optimized using binary cross-entropy with optional masking:

$$\mathcal{L}_{\text{risk}} = \sum_{j \in \mathcal{R}} m_j \cdot \mathrm{BCE}(r_j, \hat{r}_j)$$

where $m_j \in \{0, 1\}$ indicates label availability.

Binary Cross-Entropy Definition Let $y \in \{0, 1\}$ be the true label and $p = \sigma(z)$ the predicted probability. Then:

$$\mathcal{L}_{\text{BCE}}(y, p) = -\big[y \log(p) + (1 - y)\log(1 - p)\big]$$

13.3.3 Joint Optimization

The total objective combines emotion and risk losses:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{emotion}} + \sum_{j \in \mathcal{R}} \lambda_j \mathcal{L}_{\text{risk}}^{(j)}$$

where λ_j are task-weighting coefficients.

14 Inference-Time Emotion Interpretation

14.1 Motivation for Lenient Decoding

Empirical error analysis shows systematic confusion among semantically adjacent emotions:

- *Surprise* overlaps with *joy* and *fear*
- *Love* overlaps with *joy*

These confusions reflect affective proximity rather than model failure, motivating inference-time leniency.

14.2 Lenient Emotion Decoding

Let $\mathbf{p}$ denote emotion probabilities, and let $e_{(1)}, e_{(2)}, e_{(3)}$ be the top-ranked emotions.

Define compatibility sets:

$$C_{\text{surprise}} = \{\text{joy}, \text{fear}\}, \quad C_{\text{love}} = \{\text{joy}\}$$

Acceptance Rules **Surprise ambiguity:**

$$e_{(1)} \in C_{\text{surprise}} \wedge e_{(3)} = \text{surprise} \Rightarrow \hat{Y} = \{e_{(1)}, \text{surprise}\}$$

Surprise in top-2:

$$\{e_{(1)}, e_{(2)}\} \cap \{\text{surprise}\} \neq \emptyset \Rightarrow \hat{Y} = \{e_{(1)}, e_{(2)}\}$$

Love–Joy overlap:

$$\{e_{(1)}, e_{(2)}\} = \{\text{joy}, \text{love}\} \Rightarrow \hat{Y} = \{\text{joy}, \text{love}\}$$

A prediction is correct if:

$$y \in \hat{Y}$$

This converts strict multiclass evaluation into structured top-k acceptance.

14.3 Uncertainty-Aware Interpretation

Emotion classification in short-form and low-signal text frequently yields diffuse probability distributions, where multiple emotion labels receive similar confidence scores. In such cases, enforcing a single dominant emotion based on rank ordering alone may lead to over-interpretation of marginal probability differences. To address this, the system adopts an uncertainty-aware interpretation strategy at the explanation layer, while leaving the underlying model predictions unchanged.
Let $p_{(1)}$ and $p_{(2)}$ be the top two emotion probabilities, and let $\delta = 0.05$.

$$\text{EmotionState} = \begin{cases} \text{Weak} & p_{(1)} < \tau_{\min} \\ \text{Mixed} & p_{(1)} - p_{(2)} < \delta \\ \text{Dominant} & \text{otherwise} \end{cases}$$

This mechanism operates exclusively at the explanation layer and does not alter inference or evaluation.

15 Statistical Basis of Risk Thresholding

Risk classifiers produce continuous probabilities that are discretized using validation-optimized thresholds.

Let:

$$\hat{y}_i(\tau) = \begin{cases} 1 & p_i \geq \tau \\ 0 & \text{otherwise} \end{cases}$$

Optimal thresholds are selected via:

$$\tau^* = \arg\max_{\tau \in [0,1]} \text{F1}\big(\hat{y}(\tau), y\big)$$

This procedure yields category-specific thresholds reflecting class imbalance and linguistic ambiguity.

15.1 Cost-Sensitive Interpretation

Define:

$$\mathcal{R}(\tau) = C_{\text{FN}} \cdot \text{FN}(\tau) + C_{\text{FP}} \cdot \text{FP}(\tau)$$

Minimizing $\mathcal{R}(\tau)$ yields:

$$\tau^* = \arg\min_{\tau \in [0,1]} \mathcal{R}(\tau)$$

Although F1 is used operationally, the resulting thresholds implicitly encode asymmetric costs.

16 Interpretation Gates and Evidence Augmentation

16.1 Conditional Interpretation (Non-Depression)

Let p_r be the model probability for risk r; τ_r^{opt} be the learned optimal threshold; K_r be the set of risk-specific n-grams (the n-grams in this set are fixed a priori and are not learned during model training); and $c_r(x)$ be the number of matched n-grams in text x. Then the interpretation decision is:

$$\text{Interpret}(r) = \begin{cases} \text{Yes} & \text{if } p_r \geq \tau_r^{\text{high}} \\ \text{Yes} & \text{if } \tau_r^{\text{mid}} \leq p_r < \tau_r^{\text{high}} \wedge (c_r(x) \geq k_3 \vee \ell_r(x) = 1) \\ \text{Yes} & \text{if } \tau_r^{\text{low}} \leq p_r < \tau_r^{\text{mid}} \wedge (c_r(x) \geq k_2 \vee \ell_r(x) = 1) \\ \text{Yes} & \text{if } \tau_r^{\text{opt}} \leq p_r < \tau_r^{\text{low}} \wedge (c_r(x) \geq k_1 \vee \ell_r(x) = 1) \\ \text{No} & \text{if } p_r < \tau_r^{\text{opt}} \end{cases}$$

Where $\tau_r^{\text{high}} > \tau_r^{\text{mid}} > \tau_r^{\text{low}} > \tau_r^{\text{opt}}$ and:

1. $\ell_r(x) \in \{0, 1\}$ = presence of a risk-aligned subtheme
2. $\tau_r^{\text{high}} = 0.75$
3. $\tau_r^{\text{mid}} = 0.5$
4. $\tau_r^{\text{low}} = 0.3$
5. $k_1 = 6$, $k_2 = 5$, $k_3 = 3$

This is not threshold replacement. It is evidence augmentation. Optimal thresholds define the minimum admissible confidence for interpretive disclosure; keyword evidence modulates disclosure within that admissible region.

16.2 Depression as Graded Disclosure

For depression, interpretation is expressed as graded intensity rather than a binary decision to avoid categorical diagnostic framing due to the high lexical ambiguity of depressive language. The optimal threshold is set to $\tau_r^{\text{opt}} = 0.9$:

$$\text{Interpret}(r_{depression}) = \begin{cases} \text{High} & \text{if } p_r \geq \tau_r^{\text{opt}} \\ \text{Mid} & \text{if } \tau_r^{\text{mid}} \leq p_r < \tau_r^{\text{opt}} \wedge (c_r(x) \geq l_2 \vee \ell_r(x) = 1) \\ \text{Mild} & \text{if } \tau_r^{\text{mild}} \leq p_r < \tau_r^{\text{mid}} \wedge (c_r(x) \geq l_1 \vee \ell_r(x) = 1) \\ \text{None} & \text{if } p_r < \tau_r^{\text{mild}} \end{cases}$$

Where:

- $\ell_r(x) \in \{0, 1\}$ = presence of depression subtheme
- $\tau_r^{\text{mid}} = 0.7$
- $\tau_r^{\text{mild}} = 0.5$
- $l_1 = 5$, $l_2 = 3$

17 Post-hoc Grounding and Subtheme Filtering

Risk grounding is post-hoc and non-causal. Subthemes are computed via:

$$\mathrm{score}(s \mid x) \propto \sum_{e \in E_s} p(e \mid x) \cdot w_s$$

Disclosure requires:

$$s_i(x) \geq \tau_s \quad \text{with } \tau_s = 1.0$$

Lexical-gated subthemes declare:

$$\text{requires\_lexical\_evidence}(s) \in \{0, 1\}$$

$$\mathrm{score}(s) = \begin{cases} 0 & c_s(x) = 0 \wedge s \in S_{\text{lexical}} \\ \text{existing score} & \text{otherwise} \end{cases}$$

This preserves semantic precision and prevents spurious activation.

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